BRAC UNIVERSITY

Department of Computer Science and Engineering
CSE 341: Microprocessors

Examination: Quiz 1
Duration: 40 Minutes

Semester: Summer 2026
Full Marks: 20

Name:	ID:	Section:

Question 1 [6.5 marks]

PewDiePie intends to build a custom **subscriber-counter system** for his studio and is evaluating two alternative designs to achieve a **cost-effective and efficient** solution.

- **System A** is built around a **single-chip processing circuit** that requires separate chips for program storage, data storage, and peripheral interfacing.
- **System B** uses a **single integrated circuit** that can independently store the program, maintain subscriber data, read button inputs, and drive a display module.

Both systems are capable of displaying the current subscriber count, but their hardware organization differs significantly.

a. Identify the type of core device used in System A and System B. **[0.5 + 0.5 = 1]**
b. Draw the **block diagram** of **System A** and **System B**, clearly showing all major components and their interconnections. **[2 + 2 = 4]**
c. Considering PewDiePie's requirements, recommend the more suitable system and justify your choice. **[1.5]**

Question 2 [8.5 marks]

Physical Address	**71C9C**	**71C9D**	**81C9C**	**81C9D**	**61C9C**
8 bit hex data	71	CA	71	BD	71

Given, SS = 6A40h, CS = 7AC0h, DS = 8140h, ES = 6DAFh, SI = 43AEh, DI = CA3Dh.

Instruction (i): MOV BX, [DI+776Fh]

a. Deduce the physical address of the source for the **instruction-(i)**. **[1.5]**
b. Can the physical address obtained in **question-(a)** can be used as a segment number? Justify your answer. **[1.5]**
c. Determine the data store in **BX** after executing instruction-(i). **[1.5]**
d. Determine the **3rd highest** and the **lowest logical addresses** for the physical address obtained in **question-(a)**. **[4]**

Question 3 [5 marks]

a. Write **two** advantages of segmentation. **[2]**
b. A microprocessor uses **non-overlapping** segmented memory organization. The total memory capacity is **64KB**, and the maximum size of each segment is **512 bytes**. Determine the **minimum** sizes of the **segment register** and **offset register** that has to be used for this scenario. **[3]**

Question 1

a. A = microprocessor, B = microcontroller
b. Check lecture slide for block diagram of microprocessor and microcontroller
c. Microcontroller since a subscriber-counter system is relatively simple and involves repetitive task

Question 2

a. ES x 10h + (DI+776F) = 81C9Ch
b. No, since segment number in 8086 can be of maximum 16 bits.
c. 3rd highest Logical address = 81C7:002C
   Lowest Logical address = 71CA:FFFC

Question 3

a. Check slide
b. Since total memory capacity is 64KB = 2^16 Bytes, hence maximum bits required to represent Physical address = 16 bits. Since the maximum size of each segment can be 512 Bytes = 2^9 Bytes, hence the offset address can have = 9 bits. Thus minimum size of offset register = 9 bits. Therefore, minimum size of segment register = 16 - 9 = 7 bits.